

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT I Classification of Signals and Systems

Degree, Semester& Branch: III Semester B.E.ECE B

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.G.Gnana Priya

Name of the Topic: Revision of Unit I

Name of the Innovative Practice: Mind Map

Date & Duration: 15.07.2019 & 15 Minutes

Description of Mind Map Activity:

A mind map is a graphical way to represent ideas and concepts. It is a visual thinking tool that helps structuring information, helping you to better analyze, comprehend, synthesize, recall and generate new ideas. It is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

Goals(Learning Outcome):

The students will be able to classify the types of signals and systems and also analyse the various properties of signals and systems.

Use of appropriate methods:

Justification for choosing the topic using One Minute Paper Activity:

Unit I deals with the basics of signals and systems which is an important topic in concept wise and university exam point of view. So I have chosen this topic and the reason for choosing mind map activity is while recollecting the concepts and hints learnt in the classroom, the students will be able to reproduce the concept in the exam.

Effective presentation:

Details of the Implementation:

The students were randomly called and they were asked to write the topic they had learnt in the classroom as a pictorial representation in the white board.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam.

Samples images:



Reflective Critique:

- **Benefits of conducting this activity:**

The students gave oral feedback that now they got a clear idea about classification of signals and systems, and they asked me to conduct more such activities like this.

- **Challenges:**

I planned to conduct this activity within 10 minutes. But it took around 15 minutes to finish this activity, since few students got hesitate to come to the board and write, I made those students to come forward. So it took 5 minutes extra time to finish this activity.

Topic: Classification of Systems

Activity: One Minute Paper

Description of One Minute Paper:

Minute Paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. Its major advantage is that it provides rapid feedback on whether the professor's main idea and what the students perceived as the main idea are the same. Additionally, by asking students to add a question at the end, this assessment becomes an integrative task. Students must first organize their thinking to rank the major points and then decide upon a significant question.

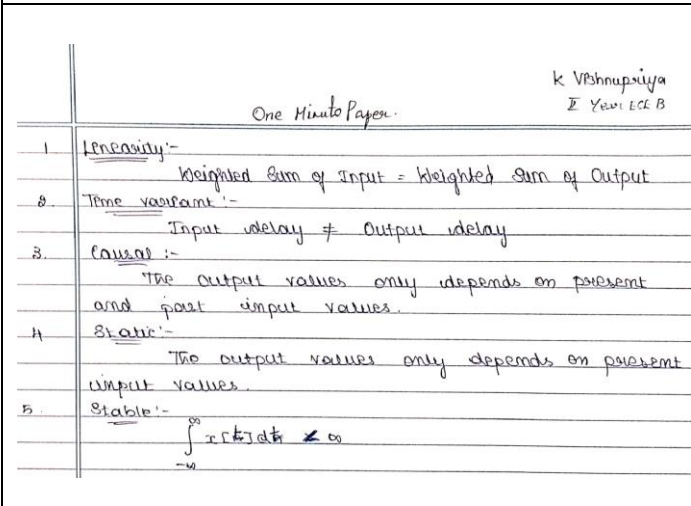
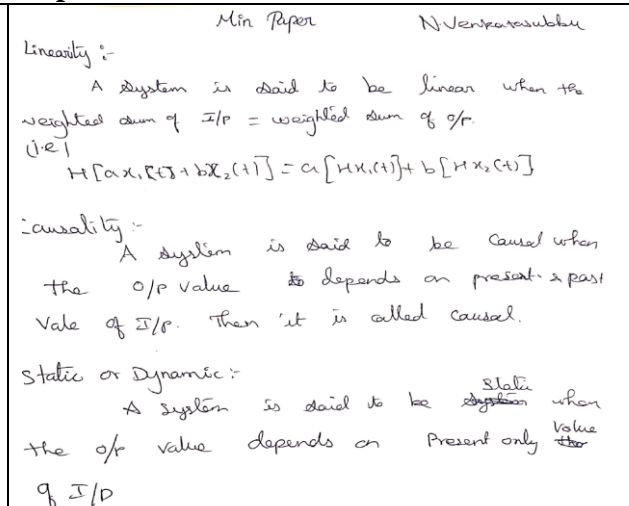
Implementation in the Classroom:

The students were asked to recall the topic learnt in the classroom and they were asked to write they have learnt during that particular class in a piece of paper.

Effectiveness of conducting the activity:

Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam. In the Internal Assessment Test-I most of the students attended the questions under this topic and scored good marks.

Samples images:

One Minute Paper	
 One Minute Paper k. Vashnupriya E Year ECE B 1. <u>Linearity</u>:- Weighted sum of Input = Weighted sum of Output 2. <u>Time variant</u>:- Input delay $\neq$ Output delay 3. <u>Causal</u>:- The output values only depends on present and past input values. 4. <u>Static</u>:- The output values only depends on present input values. 5. <u>Stable</u>:- $\int_{-\infty}^{\infty} x(t) dt < \infty$	 Min Paper N. Venkateswara Linearity:- A system is said to be linear when the weighted sum of I/P = weighted sum of O/P. (i.e.) $H[ax_1(t) + bx_2(t)] = a[Hx_1(t)] + b[Hx_2(t)]$ Causality:- A system is said to be causal when the O/P value depends on present & past value of I/P. Then it is called causal. Static or Dynamic:- A system is said to be dynamic ^{Static} when the O/P value depends on Present only the value of I/P.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	1	1			1		2		2

CO1: The students will be able to analyze the various properties of signals and systems.

References:

- <https://ocw.mit.edu/resources/res-6-007-signals-and-systems-spring-2011/lecture-notes>
- Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson, 2015.

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Innovative Practices Description

UNIT II Analysis of Continuous Time Signals

Degree, Semester & Branch: III Semester B.E.ECE B

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.G.Gnana Priya

Name of the Topic: Fourier Series Properties

Name of the Innovative Practice: Zero Minute Speech

Date & Duration: 25.07.19 & 15 minutes

Description of Zero Minute Speech:

Zero minute speech activity makes the student to come forward and share the things they have learnt during that particular class. Due to this activity the student will be able to recall the topic learnt in the classroom and they can reproduce it in the exam.

Goals(Learning Outcome):

The students will be able to represent continuous time periodic signals using Fourier series.

Use of appropriate methods:

Justification for choosing the topic using Zero Minute Speech Activity:

Usually the students won't study the properties of Fourier series and in the exam also they prefer to choose problems rather than choosing properties of Fourier Series. But if they attend properties they can score full marks when compared to solving problems because there may be a chance to do mistake while solving problems. To change the mentality of the students and to make the students attend this topic in the exam, I have chosen this topic and conducted Zero Minute Speech Activity.

Effective Presentation:

Details of Implementation:

The students were asked to recall the topic learnt in the classroom and they were asked to give a speech for few minutes about the Properties of Fourier Series. Around 7 students were randomly called and they were asked to talk about any one property.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

Due to this activity, the active learning skills of the students got increased.

Sample Images:



Reflective Critique:

- **Benefits of conducting this activity:**

By conducting this activity it was easy for the students to recollect the points learnt in the classroom. They will also be able to reproduce the concepts in the exam.

- **Challenges faced in conducting this activity:**

I planned to conduct this activity within 5 minutes. But it took around 15 minutes to conduct this activity, since few students hesitate to come to the dias and share the concept. I changed their mentality and brought them to the stage to share the concept. Only very few students got chance to come and share their ideas since the time duration for conducting this activity is limited.

Changes required for future:

Since the students feel shy to come to the dias and share the concept, next time while conducting this activity I have planned to encourage the students by giving gifts like pen, chocolates etc.,

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	2	2	1	1			1		2		2

CO2: The students will be able to apply Laplace transform and Fourier transform in signal analysis.

References:

1. Signals and Systems-Khan Academy-Visualizing the Fourier expansion of a square wave
2. Allan V.Oppenheim, S.Wilsky and S.H.Nawab, "Signals and Systems", Pearson, 2015.

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Innovative Practices Description

UNIT III Linear Time Invariant –Continuous Time Systems

Degree, Semester& Branch: III Semester B.E.ECE B

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.G.Gnana Priya

Name of the Topic: Impulse response, Convolution Integral

Name of the Innovative Practice: Quiz Dominos

Date & Duration: 20.08.19 & 20 minutes

ICT Tool Used : LCD Projector

Description of Quiz Dominos:

A quiz is a form of game or mind sport, in which the students attempt to answer questions correctly. It is a game to test their knowledge about a certain topic. This quiz dominos activity helps students practice answering questions about impulse response and convolution integral.

Goals(Learning Outcome):

The students will be able to apply the properties of impulse response and convolution integral to solve Linear Time Invariant Systems connected in series or parallel.

Use of appropriate methods:

Justification for choosing the topic using Quiz Dominos Activity:

The students feel the topic Convolution integral as tough when compared to other topics in this unit. To make the students feel easy with this topic, I conducted this activity (Quiz Dominos) for this topic which made the students to solve the problems and choose the correct answer from the given options.

Effective Presentation:

Details of the Implementation:

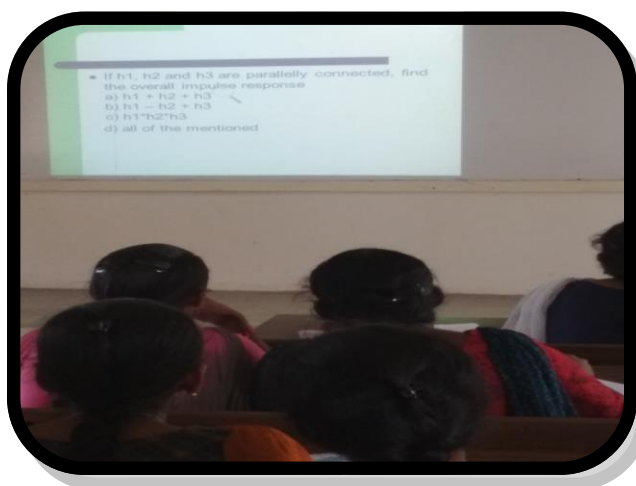
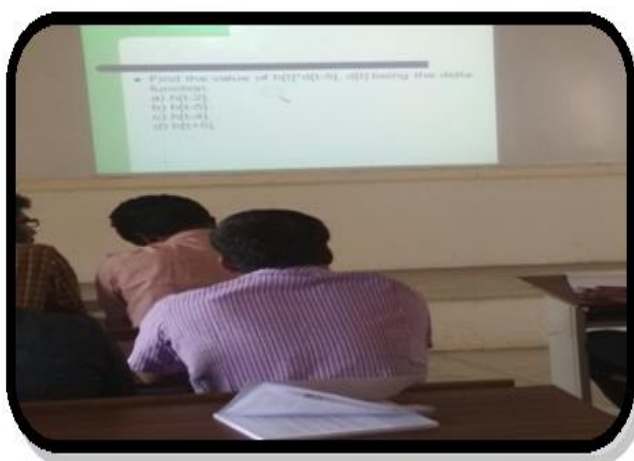
The questions were displayed in the classroom and the students were asked to answer the questions from the knowledge of what they learnt in the classroom. At the end of the activity the correct answers were discussed.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

Most of the questions were correctly answered by the students. The wrongly answered questions were discussed at the end of the class.

Sample Images:



Reflective Critique:

- **Benefits of conducting this activity:**

The students got a clear idea about this topic and they felt it was useful.

- **Challenges faced in conducting this activity:**

I planned this activity for 20 minutes and it took around 25 minutes to finish this activity, since the students took some time to solve the problems and answer.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	3	2	1	1			1		2		2

CO3: The students will be able to analyze linear time invariant continuous time systems using Laplace and Fourier Transforms.

References:

1. MIT Open Courseware/Signals and Systems/Convolution
2. Allan V.Oppenheim, S.Wilsky and S.H.Nawab, "Signals and Systems", Pearson, 2015

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Innovative Practices Description

UNIT IV Analysis of Discrete Time Signals

Degree, Semester& Branch: III Semester B.E.ECE B

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.G.Gnana Priya

Name of the Topic: Z transform

Name of the Innovative Practice: Think-Pair-Share

Date & Duration: 17.09.2019& 10 minutes

Description of Think-Pair-Share:

Think-Pair-Share is a peer-to-peer discussion strategy where students discuss their constructed responses to a prompt or question before sharing with the whole class. It is a strategy that engages every student so that checking the understanding of the whole class can be done efficiently.

Goals(Learning Outcome):

The students will be able to analyze discrete time signals using Z transform.

Use of Appropriate methods:

Justification for choosing the topic using Think-Pair-Share Activity:

Since Z transform is an important topic in concept wise and university exam point of view, I have chosen this topic and the reason for choosing Think-Pair-Share activity is the students feel free to discuss with their neighbors (friends) and then they share with the class.

Effective Presentation:

Details of Implementation:

The students were divided into pairs and they were asked to think of what they learnt in the classroom and they were asked to share with their pairs. Then one from any three pairs were asked to share the concept in front of the class.

Sample Images:





Reflective Critique:

- **Benefits of conducting this activity:**

Most of the students attended the topic Z transform in Internal Assessment Test III and scored good marks. The active learning skills of the students got increased.

- **Challenges faced in conducting this activity:**

Only very few pairs got chance to share their ideas since the time for conducting this activity is limited.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	3	2	1	1			1		2		2

CO4: The students will be able to analyze discrete time signals using DTFT and Z transform.

References:

1. Poorna Chandra S, Sasikala. B ,Digital Signal Processing, Vijay Nicole/TMH,2013.
2. Allan V.Oppenheim, S.Wilsky and S.H.Nawab, "Signals and Systems", Pearson, 2015

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Innovative Practices Description

UNIT V Linear Time Invariant - Discrete Time Systems

Degree, Semester & Branch: III Semester B.E.ECE B

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.G.Gnana Priya

Name of the Topic: Convolution Sum

Name of the Innovative Practice: Sorting Strips

Date & Duration: 27.09.2019 & 25 minutes

Description of Sorting strips:

Sorting strips deals with sharing of ideas with the group members and arranging the given strips in order that is the most important outcome of the activity. This activity increases the collaborative learning skills of the students.

Goals(Learning Outcome):

The students will be able to analyze Linear Time Invariant Discrete time systems using convolution sum.

Use of Appropriate methods:

Justification for choosing this topic using Sorting Strips Activity:

In general, students will participate eagerly in group activities instead of individual activities. In student point of view since they feel this topic as hard, I planned this sorting strips activity for this topic to make them feel easy.

Effective Presentation:

Details of Implementation:

The entire class was divided into equal groups and each group was given some pieces of paper where the procedure for doing convolution is written in each paper. Each group has to arrange the pieces of paper in correct order to perform convolution operation. The students actively participated in this activity.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

Since I conducted this activity, the collaborative learning skills of the students got increased. They actively participated in this activity.

Reflective Critique:

- **Benefits of conducting this activity:**

Usually the students don't attend this topic convolution sum since it is a lengthy procedure. But after conducting this activity they got a clear idea about this topic and most of the students attended this question asked in Internal Assessment Test III.

- **Challenges in conducting the activity:**

I planned to conduct this activity in 15 minutes. But it took around 25 minutes to complete since I mentioned numbers in each paper the students got a little confusion whether to arrange the numbers or to arrange the papers based on the sentences written in each paper. Then I clarified their doubt. So it took some 10 minutes extra time to conduct this activity.

Sample images:



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	3	2	1	1			1		2		2

CO5: The students will be able to interpret the linear time invariant discrete time systems using DTFT and Z transform.

References:

1. J.G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
2. Allan V.Oppenheim, S.Wilsky and S.H.Nawab, "Signals and Systems", Pearson, 2015